MCTE 3331

ENGINEERING DESIGN AND RELIABILITY

SEM 2 23/24

SECTION 1

Project Report :

Automatic Waste Segregation

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CHAPTER 1 : INTRODUCTION

The problem of waste management has consequently grown to be a global issue due to the ever-increasing population and increased rate of urbanization resulting in the production of an enormous amount of waste from households, industries, and institutions daily. Manual sorting of wastes is ineffective, time-wasting, cost- and labor–intensive and consequently; it is associated with high levels of imprecision and inaccuracies. It is why traditional approaches to handling wastes are becoming less useful when novel solutions to the impacts of environmental sustainability are desperately needed.

Automatic waste segregation systems are features which are far improvements to waste management. These systems employ advanced features of technology including AI, robots, and machine learning to sort through the items hence greatly lessening their reliance on humans and the consequential mistakes. Adopting these technologies in segregation of wastes assist in increasing the capacity for sorting when properly applied to the waste management process hence increasing the overall efficiency in the recycling process.

Some remarkable solutions have appeared in the past few years; this information proves the effectiveness of automated waste segregation systems. These include ZenRobotics Recycler that employs robotic arms controlled by artificial intelligence to sort construction and demolition waste to very high accuracy and in real time. The technology of computer vision and machine learning is used in AMP Robotics for sorting and recycling of recyclables and includes different materials including plastic, paper, and metals. Companies such as Sadako Technologies realized the concept of robotics by creating state-of-art robots to pick and sort the wastes at massive speed and great accuracy, hence transforming the effectiveness of waste management.

The use of such sophisticated technologies in waste management is informed by the increase in the environmental problems relating to the unproper disposal of wastes and the depletion of natural resources. Promisingly, such capabilities boost recycling and cut the use of landfill and discharge of redundant greenhouse gases into the environment as a result of efficient segregation systems. It allows recycling of useful materials from waste flows which in turn fosters the concepts of circular economy hence enhanced sustainability.

It can be understood that incorporating the instances of the automated waste segregation systems into the existing waste management frameworks is not only advantageous, but mandatory as the world progresses further towards sustainability. Such systems provide a plausible solution to the difficulties caused by approaches to waste segregation, providing the grounds for a new age in the improvement of waste management procedures. As AI, robotics, and machine learning continue to evolve, it will present more opportunities for stretching the field further and coming up with efficient ways of improving the effectiveness and sustainability of the segregation of wastes.

CHAPTER 2 : PROBLEM DEFINITION AND IDENTIFICATION

2.1 Product Literature

Automated methods of waste disposal have been developed in recent years and several products have been designed to work through mechanisms of automated waste segregation with certain characteristics. These products include AI, robotics, machine learning among others to improve the speed and effectiveness of sorting out the wastes. Besides, the usage of these technologies in the formation of the products serves to minimize time and effort while enhancing the efficiency of environmental control.

This section provides an overview of three prominent solutions: such as the ZenRobotics Recycler, AMP Robotics, and Sadako Technologies. All these products are individual masterpieces aimed at responding to definite difficulties of waste segregation and adapted to provide the highest possible level of differing features in waste management. Further down each of these products is described in terms of its dimensions, weight, sorting capability, power consumption, as well as the run time.

ZenRobotics Recycler

Dimension	**2000 x 3000 x 2500 mm**
Weight	**800 kg**
Sorting Capacity	**50 tons per hour**
Power Consumption	**15 kW**
Run Time	**Continuous operation with regular maintenance**

ZenRobotics Recycler is a cutting-edge robotic waste sorting, which helps to sort construction and demolition waste with the help of AI-controlling robotic arms. This system is generic to accommodate different types of materials; this makes it easier to sort the material with a high degree of accuracy and considerably minimises the amount of work done by hand. And it runs with monitoring and analyzing in real-time and is easily diagnosable when there is a problem with performance.

AMP Robotics

Dimension	**1800 x 2500 x 2200 mm**
Weight	**650 kg**
Sorting Capacity	**40 tons per hour**
Power Consumption	**12 kW**
Run Time	**Continuous operation with regular maintenance**

Specifically, AMP Robotics utilizes computer vision and AI to identify, sort, and recycle some elements of items such as containers, cans, and bottles. Due to the versatility of the system, it can accept a variety of materials; for instance, plastics, papers and metals among others. In its operation, AMP Robotics can keep on enhancing its sorting capacities, thus, regarding its leading position in waste management solutions, it can learn and adapt.

Sadako Technologies

Dimension	**1600 x 2300 x 2100 mm**
Weight	**600 kg**
Sorting Capacity	**35 tons per hour**
Power Consumption	**10 kW**
Run Time	**Continuous operation with regular maintenance**

Currently, Sadako Technologies have designed robots with artificial intelligence sense that enables the robots to pick and sort the waste materials with immense efficiency and precision. They play a major role of improving on the efficiency of waste management and save a lot of time and efforts that would otherwise be required in sorting of wastes and boosting the throughputs in recycling processes. This is a good addition to any waste management facility since the operations of the system are very fast and accurate.

2.2 Technical literature

To enable the Automatic Component Segregator, it is first necessary to use a smart transistor testing kit that cuts and reassembles parts of the circuit. Technical literature is the paperwork that contains the detailed information of the product, the design, the work and the care. This can include general guides and instructions currently popular among users, installation manuals, descriptions of the technical characteristics of the product, instructions in case of the product's malfunction, and safety standards. Technical literature is employed to inform the user about the capabilities and liabilities of the product as well as presenting to the user the information required to correctly install, set up and maintain the product.

Automatic Component Segregator Enabled with a Smart Transistor Testing Kit

The name ACE with STTK refers to another creative invention of the researchers that is an apparatus intended for the automatic segregation of the functional and non-functional components from the waste PCBs with an insignificant amount of human interference. Described by Nitin et al. and published electronically on June 16, 2017, the patent named 'Sorting and function check method for e-waste and apparatus thereof' (TW 201720535 A), the invention is characterized by multiple functional sorting stages.

Product Design and Function: Mechanical, electrical, and software components that are interrelated make up the apparatus with a view of accomplishing its functions. The waste PCBs are first introduced into the system for a primary classification aimed at excluding large elements. After this, a smart transistor testing kit checks the statute of individual components. Complex components are arranged for possible reuse, simple ones are gathered for recycling or for being disposed of. It has been designed in such a manner that there will be lesser use of manpower and at the same time the efficiency of the recovery rate of the material will be very high.

Safety Information: Precautions that are required in order to give safe operations are called for by the specific use of the device. It must operate in a contained environment so as not to expose workers and others to dangers that come with e-waste. Personal protective equipment like gloves and safety goggles should be worn by the operators to avoid cases of cut or exposure to chemicals. It is very important for the system to be checked at some set intervals, so the current state of the equipment used can be identified and possible dangers from these flaws avoided.

Error alerts from the transistor testing kit indicate issues with component functionality tests; users should recheck the positioning and connectivity of the PCBs within the system.
Technical Specifications: Dimensions: 2000 x 3000 x 2500 mm Weight: 800 kg Sorting Capacity: 50 tons per hour Power Consumption: 15 kW Run Time: Continuous operation with regular maintenance Environmental Protection: Given the presence of hazardous materials in

e-waste, proper disposal and recycling are crucial. Operators should follow local regulations and guidelines for disposing of non-functional components and materials that cannot be recycled.

2.3 Patent Literature

2.3.1 Device for waste segregation and collection of segregated waste
Author : Marcin et al.
Published : 15 July 2021
Patent Number : US 2021/0214154 A1

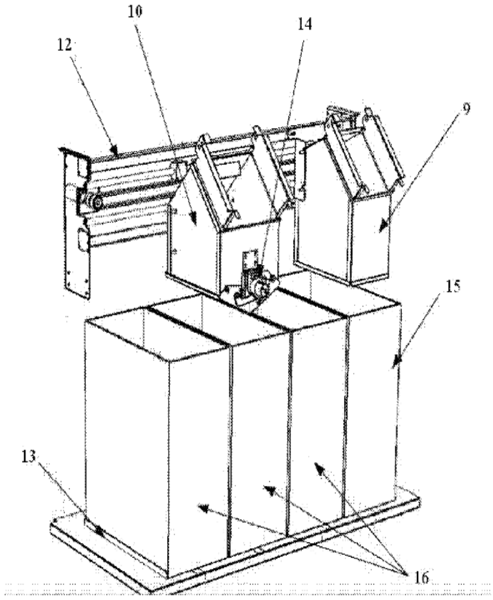


fig. 4

This patent consists of a device for waste segregation and collection of segregated waste, featuring a waste recognition, segregation, and distribution section, a compression mechanism, and storage bins. It is distinguished by the storage bins (15)(16), which have fill level sensors and are inserted into the guideways of the base (13). Directly over one of the side storage bins (15), there is a throwing box attached to the casing, with a throwing tunnel (9) and a lid with a sensor that detects whether the lid is open or closed (4). Above the bins (16), on a system of horizontal guideways (12) attached to the casing, there is a distribution cart (10) that moves sideways on the guideways, powered by a gear motor controlled by a control system. The distribution cart (10) includes a throwing mechanism (14) with at least one automatic throwing lid (11). The distribution cart operates with its own throwing box, which also has a lid with a sensor that detects whether it is closed or open (6). Within the throwing channel of this throwing box (6), there is a sensor of the waste recognition system—a camera.

2.3.2 Automatic component segregator enabled with a smart transistor testing kit(s)
Author : Nitin et al.
Published : 16 June 2017
Patent Number : TW 201720535 A

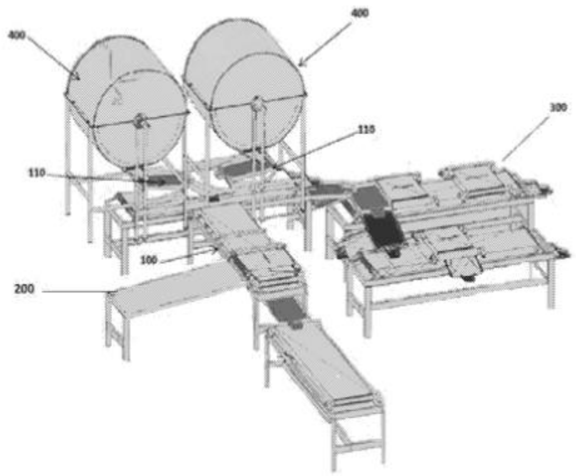
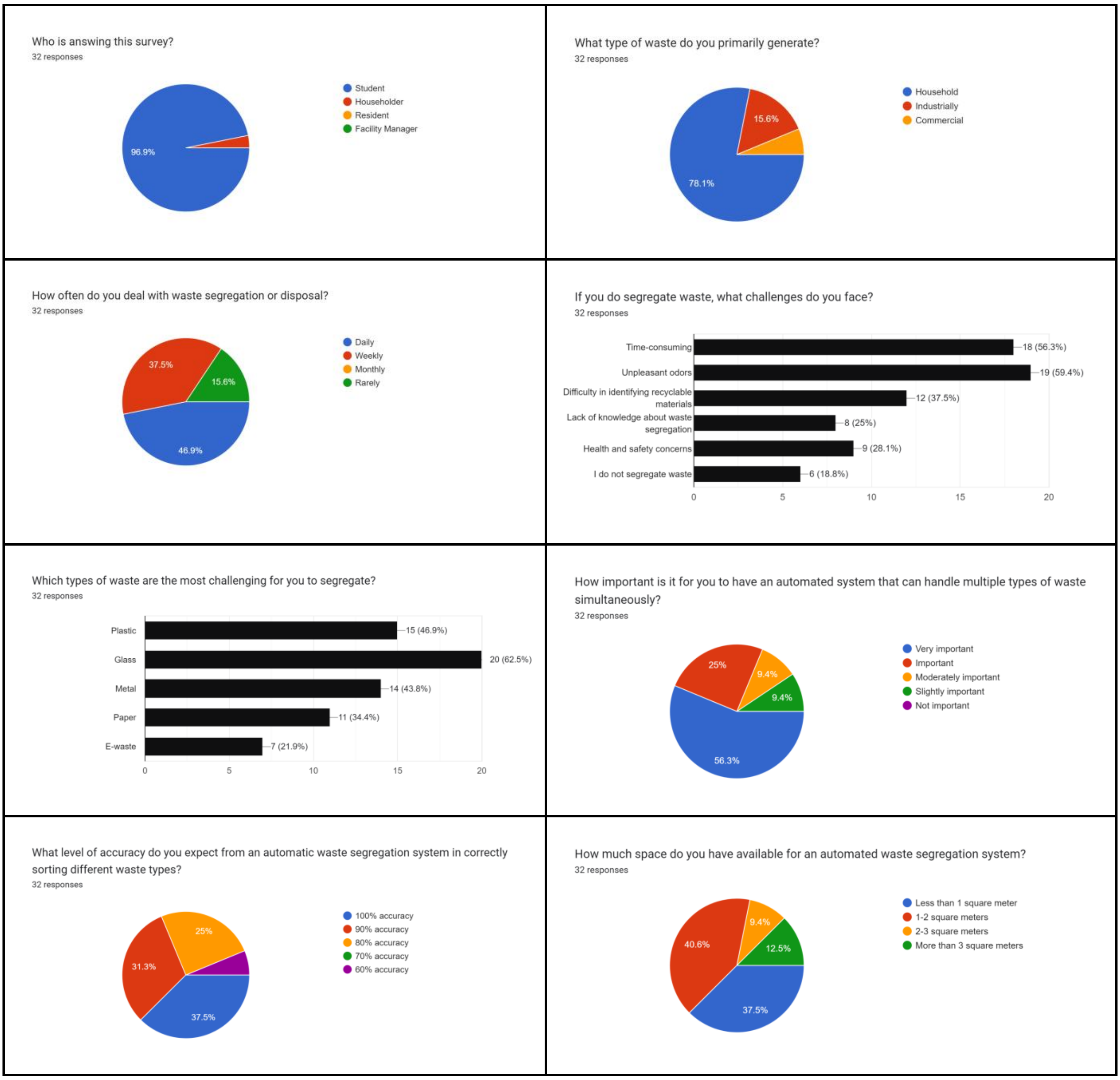


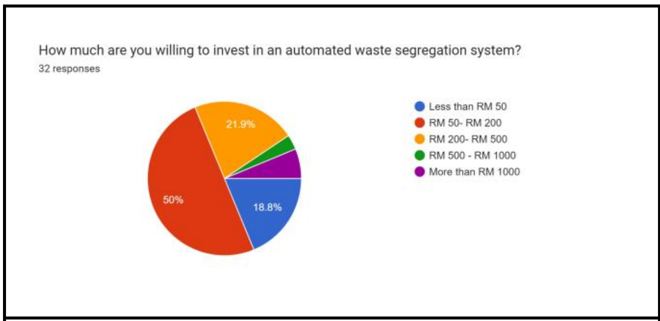
Fig. 1

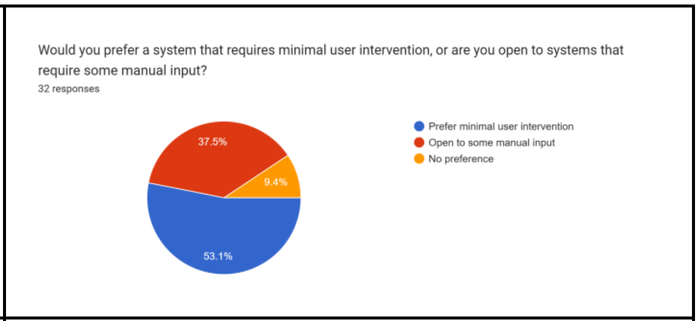
The present invention pertains to an apparatus designed for the automatic segregation of functional and non-functional components from waste printed circuit boards or electronic waste with minimal or no manual intervention. The apparatus includes a component segregation section and a functionality check section, which is equipped with at least one transistor testing kit to distinguish between functional and non-functional transistors at various stages based on their functionality, nature, and reusability. The components from the waste product are then separated through multiple stages, involving steps such as shredding, size sorting, metal sorting, and the sorting of reusable parts such as chips, transistors, resistors, etc. In the functionality check section, the extracted transistors undergo a physical integrity inspection followed by a conductivity test. If a transistor passes both the physical integrity and conductivity tests, it is deemed suitable for reuse. Otherwise, it is directed to mechanical recycling.

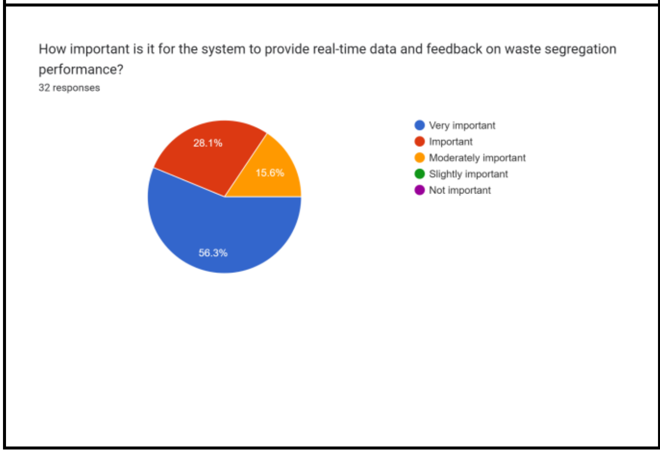
2.4 Customer Survey

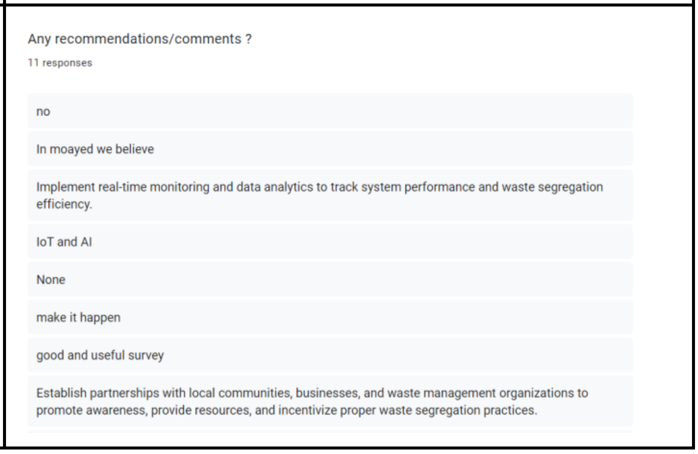
In the context of product development, customer satisfaction is paramount, as customer preferences and feedback significantly influence design choices and enhancements. To ensure that our automated waste segregation system meets the specific needs and expectations of its users, we conducted a comprehensive customer survey targeting a broad audience, including IIUM students, residents, and various other stakeholders. This survey aimed to gather direct insights from the target audience regarding their attitudes, knowledge, and behaviors toward waste management and segregation.










Any recommendations/comments ?

11 responses

no

In moayed we believe

Implement real-time monitoring and data analytics to track system performance and waste segregation efficiency.

IoT and AI

None

make it happen

good and useful survey

Establish partnerships with local communities, businesses, and waste management organizations to promote awareness, provide resources, and incentivize proper waste segregation practices.

The survey was distributed through various digital channels, including WhatsApp, Telegram, and social media platforms, utilizing Google Forms to facilitate easy and accessible participation. A total of 32 people participated in the survey, providing valuable data that helps us understand the critical aspects and features desired in an automated waste segregation system. The gathered responses offer a detailed view of the current challenges faced in waste management, the importance placed on different system attributes, and the preferences related to accuracy, intervention, and budget.

Most of the people who answered this survey are students which has a percentage of 91.6 percent. The average of these people deal with waste segregation or disposal daily and most of these people dispose of the waste in their household. Unpleasant odors are the biggest challenges faced by people while glass is the most challenging type of waste when dealing with waste segregation. 56.3 percent of people feel that it is very important to have an automated system that can handle multiple types of waste simultaneously. Average of the person would like to have an automatic waste segregation system that has 100% accuracy, a space of 1-2 m, and also prefers to have minimal user intervention. Lastly, the average of people are willing to invest at least RM 50- RM 200 for this automatic waste segregation machine.

2.5 Problem Statement

Public places need an efficient and effective way to manage and segregate waste because improper waste segregation in high-traffic areas leads to increased environmental pollution, decreased recycling efficiency, and higher waste management costs. Manual sorting of waste in public spaces is often labour-intensive, prone to errors, and can be unhygienic for workers. An automated waste segregation system designed for public use can address these issues by accurately sorting recyclable materials like metals, plastics, glass, and paper, thereby promoting sustainable waste management practices and reducing the operational burden on public maintenance services.

CHAPTER 3 : CONCEPTUAL DESIGN

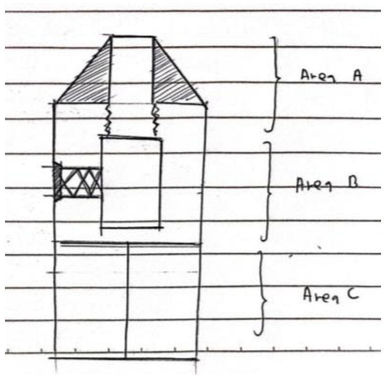
3.1 Objectives

The objectives of this project are :

1. To design a machine that can segregate recycled waste (Paper, glass, plastic, and metal) to minimize the manual efforts of sorting.
2. To develop prototype of the machine
3. To evaluate performance of the machine

3.2 Concept of the system

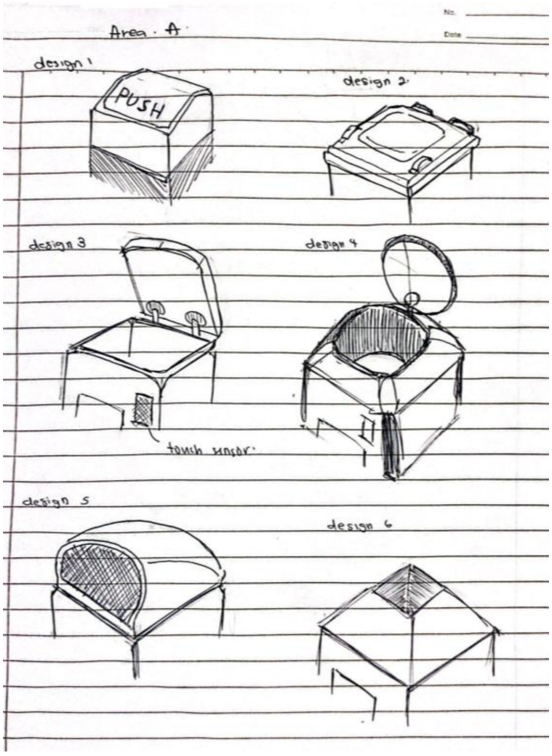
After conducting research to find a suitable solution to the problem outlined in the problem statement, our group developed three proposed combination designs based on a few designs. First of all, The design of the automatic waste segregation system is structured into three distinct areas: Area A (the head), Area B (the body), and Area C (the bottom). Each area serves a specific function within the system.



Concept Image 1.0

Area A (the head) : This section acts as the entry point of the bin, functioning as a collector for incoming waste. It is designed to facilitate the initial collection of materials.

Briefly explanation about designs for Area A ;
Design A1 : Manual push lid
Design A2 : Manual pull lid
Design A3 : Automatic open lid by pressing touch sensor and square shape
Design A4 : Automatic open lid by pressing touch sensor and circle shape
Design A5 : Open lid
Design A6 : Open lid and pyramid shape



Concept Image 1.1

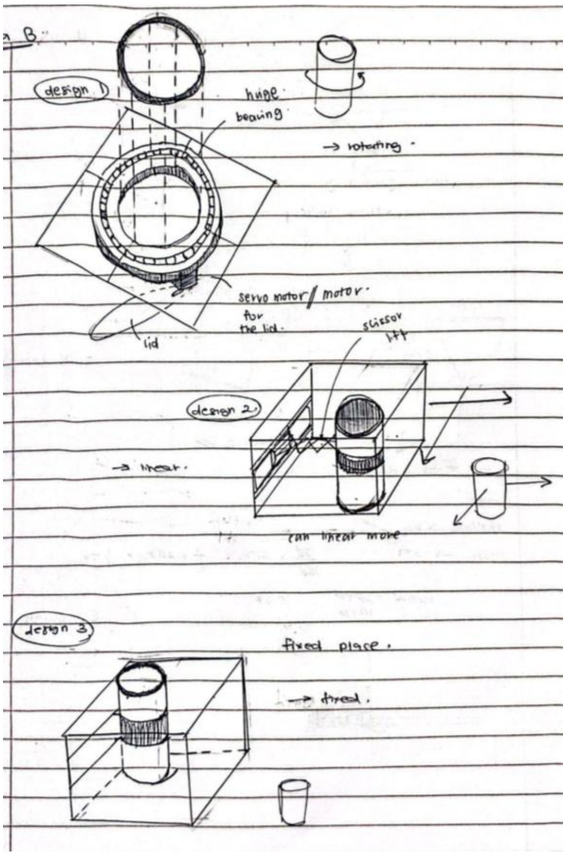
<u>Area B (the body)</u>: This section is dedicated to the detection and sorting of waste in a cylinder. Equipped with a camera and AI technology, Area B accurately recognizes and categorizes the materials. This sorting mechanism ensures that each type of waste is correctly identified.

Briefly explanation about designs for Area B ;

Design B1 : There will be a huge bearing at the bottom of Area B for a rotation and attached with a servo motor to control (open and close) a piece of metal (circle shape) at the bottom of Area B. After the inspection, it will rotate to align with its compartment , then the servo will activate.

Design B2 : The concept is the same as a 3d printer, it can move in a direction of x and y axis. After the inspection, it will move linearly to its compartment and the servo will activate.

Design B3 : The cylinder is not moving (fixed position).



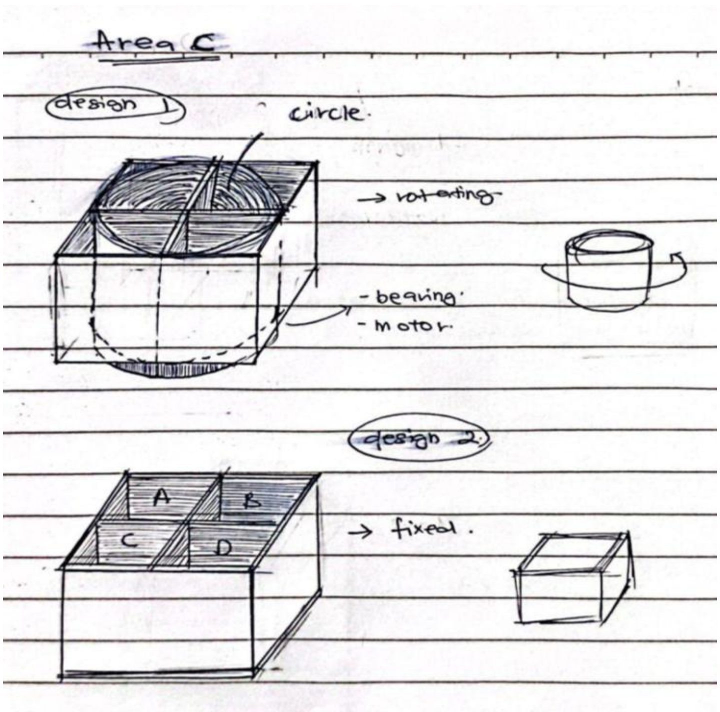
Concept image 1.2

<u>Area C (the bottom):</u> This section serves as the storage compartment for the sorted materials. It consists of four distinct compartments designated for metal, plastic, glass, and paper. This organisation allows for efficient separation and storage of recyclable waste.

Briefly explanation about designs for Area C ;

Design C1 : In this design, this part rotates accordingly, it will use a bearing and motor at the bottom of Area C.

Design C2 : The compartment is not moving (fixed position).



Concept image 1.3

After brainstorming various ideas, we proposed several combinations of designs for Areas A, B, and C. The proposed combinations are as follows:

First Combination (**Design 1**) :
- Area B: Design 2 (linear movement mechanism similar to a 3D printer)
- Area C: Design 2 (fixed compartments)

Second Combination (**Design 2**):
- Area B: Design 3 (fixed cylinder)
- Area C: Design 1 (rotating compartments with bearing and motor)

Third Combination (**Design 3**):
- Area B: Design 1 (rotating mechanism with servo motor)
- Area C: Design 2 (fixed compartments)

After further evaluation, we decided to select the best design using the Pugh Chart method. In this method, each design is compared to a reference design (Design 2) based on specific crucial criteria.

3.2.1 Pugh Chart

Selection Criteria	Concepts		
	Design 2	Design 1	Design 3
Cost	DATUM	S	-
Ease of operation		+	S
Speed of operation		+	S
Does not Jammed		S	S
Consumption energy		S	S
Weatherproof		S	S
Dimension		S	S
Sensor Accuracy		S	S
Weight		+	+
Durability		S	S
Safety		S	S
Reliability		+	+
Sum (+)		4	2
Sum (-)		0	1
Sum		4	1

The Pugh Chart above clearly demonstrates that Design 1 is the optimal choice, as it has the highest number of positive evaluations.

3.3 Engineering Characteristics

No	Engineering characteristics	Unit	Improvement Direction	Explanation
1	Jamming probability	%	↓	A low jamming probability keeps the system running smoothly and minimizes downtime for maintenance.
2	Time to run the machine	s	↓	This should be as short as possible to improve efficiency.
3	Sensor accuracy	n/a	↑	This should be high to ensure that waste is sorted correctly.
4	Complexity of the system	n/a	↓	This should be relatively low to make the system easier to maintain and repair.
5	Material Rigidity	Mpa	↑	the materials used should be strong enough to withstand the weight and impact of waste items.
6	Weight	kg	↓	This should be minimized to reduce costs and improve portability.
7	Size	m^2	↓	A smaller system will be more affordable and easier to move
8	Material	Mpa	↑	The materials used in your

	toughness			system should be tough enough to withstand wear and tear.
9	Weather resistance	n/a	↑	If the system will be used outdoors, the materials used should be weather-resistant.

3.4 Quality of deployment - House of Quality (HOQ)

Improvement Direction		↓	↓	↑	↓	↑	↓	↓	↑	↑
Units		%	s	n/a	n/a	Mpa	Kg	m^2	Mpa	n/a
House Of Quality / Customer Requirements	Importance Weight Factor	Jamming Probability	Time to run the machine	Sensor Accuracy	Complexity of the system	Material Rigidity	Weight	Size	Material Toughness	Weather resistance
Accuracy of segregation	5	9		9		3				
Ease of use	5	9	3		9		1	1		
Speed of operation	5	9	9	9	1					
Tool-less Installation	2					1	3	9		
Maintenance requirement	3				9				1	
Capacity	4						3	9		
Five-Year lifetime	4	3			1	3			9	9
Cost	2				3					
Weatherproof	4					1			3	9
Aesthetic Design	2				1	1		3		
Does Not Jam	5	9	9	9						3
Raw Score	Total 782	192	105	135	89	35	23	65	51	87
Relative Weight%		24.55	13.42	17.26	11.38	4.48	2.94	8.31	6.52	11.13
Rank Order		1	3	2	4	8	9	6	7	5

Based on the figure above, the House of Quality (HOQ) clearly ranks the engineering characteristics according to customer requirements. By analyzing the HOQ, we can identify and prioritize what our customers expect from our product. It is found that the probability of jamming is the primary concern for our future customers. Lowering the jamming probability will ensure a higher degree of reliability in our product. This is followed by sensor accuracy, which is ranked second and is crucial due to its strong connection to the safety features of our product. The time required to run the machine ranks third, highlighting its importance as the machine needs to be portable.

3.5 Product Design Specification

3.5.1 Product Overview

- **Name:** Automated Waste Segregation System

- **Function:** To automatically segregate household and commercial waste into categories such as metal, plastic, paper and glass, enhancing recycling efficiency and reducing the need for manual sorting.
- **Dimensions:**
  - 600mm x 600mm x 1000mm
- **Material:** Durable steel and high-quality plastic

3.5.2 System Components

I. Waste Identification System

The Waste Identification System employs cutting-edge sensors such as optical, infrared, and inductive sensors, coupled with AI algorithms, to precisely categorize various types of waste. It achieves a minimum accuracy of 90% in waste identification, ensuring efficient sorting processes. Regular calibration and sensor cleaning are essential maintenance tasks to uphold accuracy over time, ensuring reliable performance in waste segregation operations.

II. Sorting Mechanism

The Automated Waste Segregation System features a mechanical setup with a linear motion of a waste sorting mechanism to automatically separate waste into specific bins. It efficiently handles up to 5 kg of mixed waste in just 10 minutes. Built from durable, low-maintenance materials, the system ensures reliability and longevity in waste management operations.

III. User Interface

The user interface features a touchscreen for operating and monitoring the system. It includes an intuitive menu system for easy user settings and real-time monitoring. Enhanced usability is ensured with multiple language options and visual aids.

IV. Security Features

The system incorporates safety features such as sensors to prevent accidents and unauthorized access. An easily accessible emergency stop button is also provided to swiftly halt operations in critical situations, ensuring user safety and system integrity.

V. Notification System

The notification system provides real-time alerts to users via a GSM module, notifying them of important events such as system errors, maintenance requirements, and full waste bins. Additionally, it enables remote monitoring and control through a dedicated mobile app, enhancing operational efficiency and user convenience.

VI. Microcontroller

The microcontroller serves as the central brain of the system, utilizing an Arduino microcontroller to seamlessly integrate and control all components. It operates with a lithium battery backup, ensuring uninterrupted functionality during power outages. The centralized control code manages critical functions such as the door system, notification alerts, sorting mechanism, and user interface, optimizing system performance and reliability.

VII. RFID Scanner

The RFID scanner plays a crucial role in user authentication for the waste segregation system. It identifies and verifies users via RFID technology, restricting access solely to authorized individuals. This enhances overall security by ensuring that only registered users can interact with the system, safeguarding against unauthorized use and promoting efficient waste management practices.

VIII. Keypad

The keypad serves as a backup access method for the waste segregation system, particularly useful when the RFID scanner encounters issues. It enables users to input a unique code for accessing their waste segregation status or interacting with the system. This functionality ensures continuous usability and reliability, even in the event of RFID scanner malfunctions or failures.

3.6 Gantt Chart 1

Automatic Waste Segregation

Process	2	3	4	5	6	7	8	9	10	11	12	13	14
Problem Statement	■												
Project Selection		■											
Data Collection			■	■	■								
Conceptual Design						■	■						
Embodiment Design								■	■	■			
Detail Design											■	■	
Reporting and Submitting Report													■

CHAPTER 4: MODELLING OF THE SYSTEM

4.1 Mechanical Subsystem

4.1.1 Sorting Mechanism Calculation:

In our project, we are using linear actuators or stepper motors to sort the waste effectively and accurately. We use the similar concept movement of 3D Printing which is the actuators that we used will move sorting arms horizontally and vertically across the sorting area. The sorting mechanism will accurately direct waste items into designated bins based on their material types. Given their accuracy and ability to support a substantial load, linear actuators are the ideal option for our movement mechanism. Additionally, they can efficiently manage the weight of waste objects and the movement of the sorting arm, further reinforcing their suitability.

As we use linear actuators or stepper motors for the movement of the sorting mechanism, we need to find safety factors to ensure that the stepper motor we select has sufficient torque to move the sorting arms of the automated waste segregation system. This ensures that the motor can handle the load of the waste and the mechanical structure without failing or stalling.

1. Determine the Load

- Weight of the sorting arm (W) : 0.5 kg (maximum weight)
- Force due to weight (F): $F = m \times g$

  Where :

$$m = 0.5kg$$

$$g = 9.81m/s2$$

$$F = 0.5kg \times 9.81m/s2 = 4.905N$$

2. Identify the Lever Arm Length (r)

- Lever arm length (r): 0.476 meters

3. Calculate the Torque Required (τ)

- Torque (τ) is calculated using the formula:

$$\tau = F \times r$$

Where:

- Force (F): 4.905 N
- Lever arm length (r): 0.476 meters

$$\tau = 4.905N \times 0.476m = 2.334N \cdot m$$

A typical NEMA 23 stepper motor has a holding torque of around 1.26 N·m to 3 N·m, depending on the model and manufacturer. For this calculation, we will consider a NEMA 23 motor with a holding torque of 2.5 N·m.

Comparison

- Required Torque: 2.334 N·m
- Holding Torque of NEMA 23: 2.5 N·m So, from this calculation, we obtain that the motor of NEMA 23 can sustain the torque required.

4.2 Electrical Subsystem

4.2.1 Power Requirements:

1. Microcontroller (Arduino Mega 2560)
    a. Voltage: 5V
    b. Current: 50mA
    c. Power: $P = V \times I = 5V \times 0.05A = 0.25W$

2. Stepper Motors (NEMA 23) - 2 motors
    a. Voltage: 12V
    b. Current: 2.8A per phase
    c. Power per motor: $P = V \times I = 12V \times 2.8A = 33.6W$
    d. Total power: $33.6W \times 2 = 67.2W$

3. Servo Motor (MG996R)
    a. Voltage: 6V
    b. Current: 1.5A (assuming peak current)
    c. Power: $P = V \times I = 6V \times 1.5A = 9W$

4. Ultrasonic Sensor (HC-SR04) - 2 sensors
    a. Voltage: 5V
    b. Current: 15mA per sensor
    c. Power per sensor: $P = V \times I = 5V \times 0.015A = 0.075W$
    d. Total power: $0.075W \times 2 = 0.15W$

5. WiFi Module (ESP8266)
    a. Voltage: 3.3V
    b. Current: 170mA
    c. Power: $P = V \times I = 3.3V \times 0.17A = 0.561W$

6. LEDs - 3 LEDs
    a. Voltage: 2V
    b. Current: 20mA per LED

c. Power per LED: $P = V \times I = 2V \times 0.02A = 0.04W$
d. Total power: $0.04W \times 3 = 0.12W$

7. LCD (16x2)
    a. Voltage: 5V
    b. Current: 2mA
    c. Power: $P = V \times I = 5V \times 0.002A = 0.01W$

8. Camera (Raspberry Pi Camera Module V2)
    a. Voltage: 5V
    b. Current: 250mA
    c. Power: $P = V \times I = 5V \times 0.25A = 1.25W$

9. Touch Sensor (TTP223)
    a. Voltage: 5V
    b. Current: 15mA
    c. Power: $P = V \times I = 5V \times 0.015A = 0.075W$

10. Power Adapter
    a. Input: 100-240V AC, 50/60Hz
    b. Output: 12V DC, 2A
    c. Power: $P = V \times I = 12V \times 2A = 24W$

4.2.2 Power Consumption and Battery Sizing

Total Power Consumption

- Microcontroller: $0.25W$
- Stepper Motors: $67.2W$
- Servo Motor: $9W$
- Ultrasonic Sensors: $0.15W$
- WiFi Module: $0.561W$
- LEDs: $0.12W$
- LCD: $0.01W$
- Camera: $1.25W$
- Touch Sensor: $0.075W$

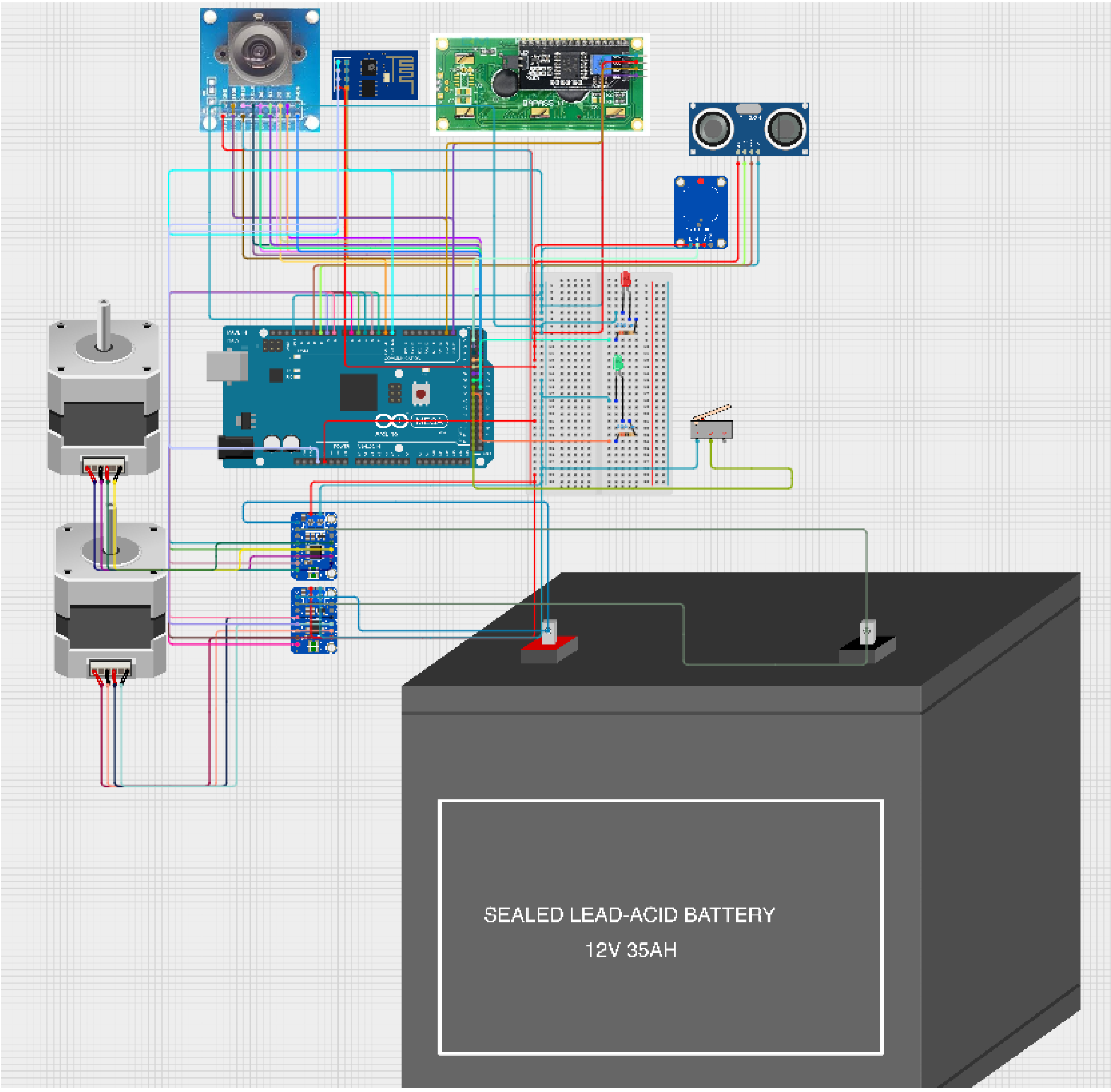
$$Ptotal = 78.616W$$

So, the power supply capacity needs at least 80W capacity (12V, 6.67A).

Battery Capacity for Desired Backup Time:
For 2 hours: $Capacity\ (Wh) = 78.616W \times 2hours = 157.232Wh$

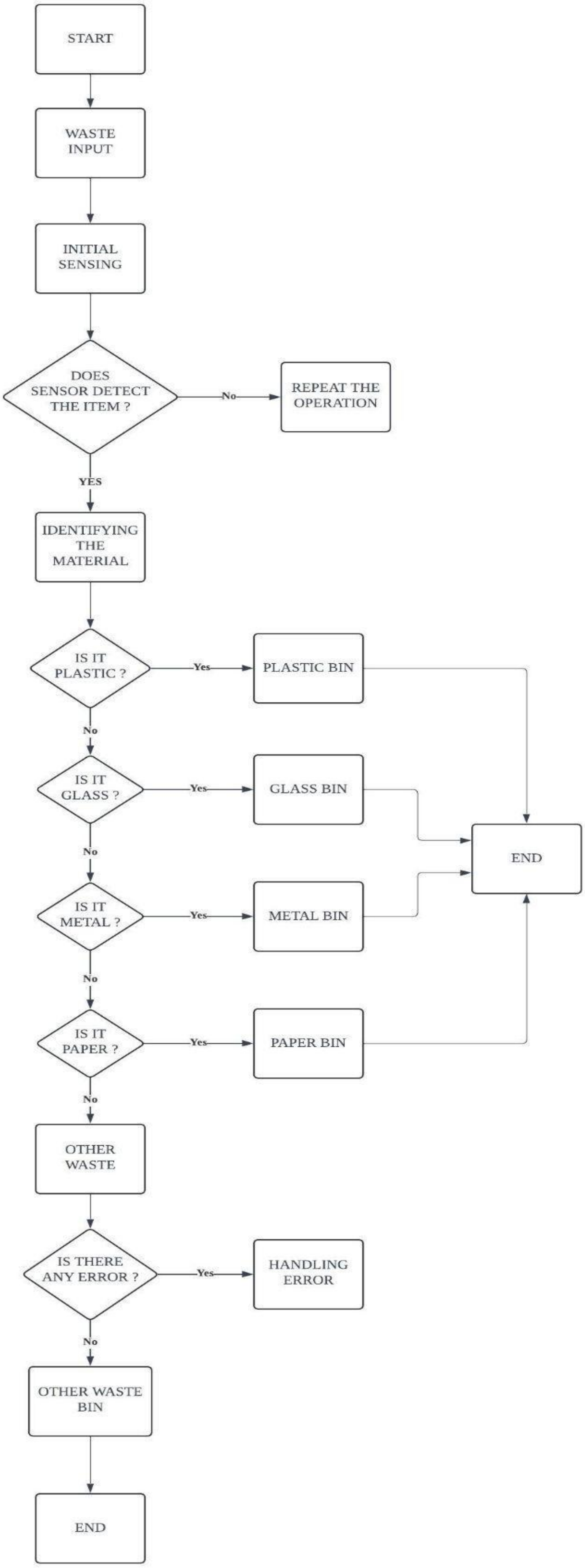
At 12V: $Capacity\ (Ah) = 157.232Wh/12V = 13.1Ah$

4.2.3 Circuit Diagram



4.3 Software and Algorithm

Arduino Mega will be used for this project. Hence, the coding will be using C++ language. The algorithm of the code:



CHAPTER 5 : DESIGN OF THE SYSTEM

5.1 Component Design

NO	COMPONENT	SPECIFICATION
1	Microcontroller	● **Model** : Arduino Mega 2560 ● Processor : ATmega2560 ● Digital I/O Pins : 54 (15 PWM outputs) ● Analog Input Pins : 16 ● Operating Voltage : 5V ● Flash Memory : 256 KB ● SRAM : 8 KB
2	Motor driver	● **Model :** DRV 8825 Stepper Motor Driver ● Voltage : 8.2V to 45V ● Current : 1.5A per phase (2.2A peak) ● Microstepping : Up to 1/32 step ● Interface : Step and direction control
3	Bipolar Stepper Motor	● **Model :** NEMA 17 ● Step Angle: 1.8° per step ● Holding Torque: 45 N·cm (63.74 oz·in) ● Current: 1.5A per phase ● Voltage: 12V
4	Servo Motor	● **Model :** MG996R ● Operating Voltage: 4.8V - 7.2V ● Stall Torque: 9.4 kg·cm (4.8V), 11 kg·cm (6V) ● Speed: 0.19 sec/60° (4.8V), 0.14 sec/60° (6V) ● Rotation: 180°
5	Ultrasonic sensor	● **Model :** HC-SR04 ● Operating Voltage: 5V ● Measuring Range: 2cm - 400cm ● Resolution: 0.3cm ● Angle: 15 degrees
6	Limit switch	● **Model :** Microswitch ● Operating Voltage: 5V - 250V ● Current: 5A ● Contact Type: SPDT (Single Pole Double Throw)

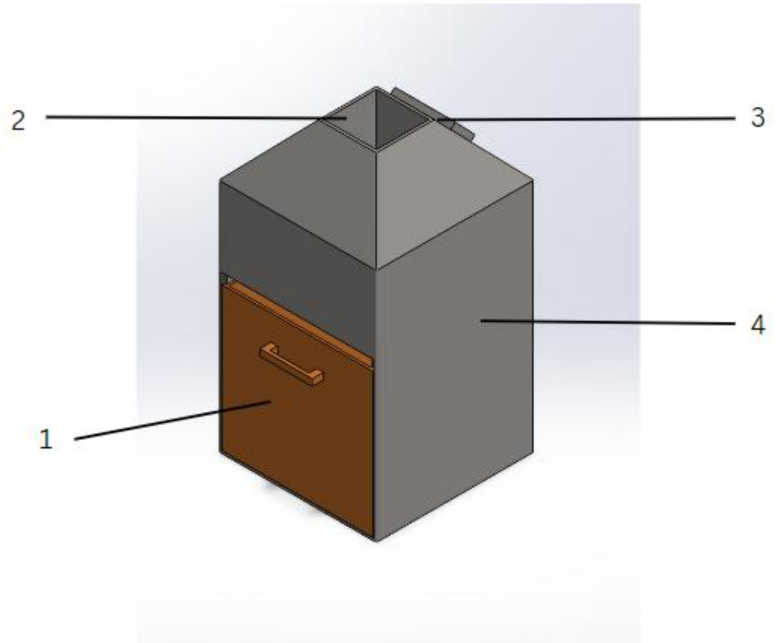
7	Wifi module	• **Model :** ESP8266 • Operating Voltage: 3.3V • Wi-Fi Standard: 802.11 b/g/n • Frequency Range: 2.4GHz - 2.5GHz • Flash Memory: 4 MB
8	Led	• **Model :** 5mm LED • Operating Voltage: 2V - 3V • Current: 20mA • Colours: Red, Green, Blue (RGB option available)
9	LCD	• **Model :** 16x2 LCD (HD44780) • Operating Voltage: 5V • Interface: Parallel • Backlight: LED backlight • Character Size: 5x8 dots
10	Wire	• **Type:** Black Red Wire • Length: 20m • Gauge: 22 AWG
11	Frame	• **Material :** Stainless Steel or Durable Plastic • 600mm x 600mm x 1000mm (cuboid shape) • Thickness : 30mm x 30mm • The whole main body
12	Breadboard	• Size: 830 tie-points • Dimensions: 16.5 x 5.4 x 0.85 cm • Rows: 2 power rails, 63 rows • Columns: 10
13	Camera	• **Model :** Raspberry Pi Camera Module V2 • Resolution: 8 Megapixels • Video: 1080p at 30fps • Interface: CSI connector
14	Power Adapter	• Input: 100-240V AC, 50/60Hz • Output: 12V DC, 2A • Connector: Barrel Jack
15	Lids	• **Material:** Stainless Steel or Durable Plastic • Shape : Square • Automatic with Touch Sensor
16	Touch Sensor	• **Model:** TTP223 Capacitive Touch

		Sensor Module ● Operating Voltage: 2.0V to 5.5V ● Output: Digital High/Low
17	Battery Backup	● **Model:** Li-Ion Rechargeable Battery Pack ● Voltage: 12V ● Capacity: 2000mAh ● Output Current: 1A
18	Microcontroller shield	● **Model :** Arduino Mega Shield for stepper motor drivers (CNC Shield V3) ● Compatible with Arduino Mega 2560 ● Designed for easy connection of multiple stepper motor drivers, such as the DRV8825 ● Voltage Range: Typically 12-36V DC ● Jumpers to set microstepping resolution (up to 1/32 step with DRV8825) ● Supports up to 4 stepper motors (X, Y, Z, and an additional axis)
19	Bearing	● **Model :** 608ZZ ● Inner diameter : 8mm ● Outer Diameter (OD): 22mm ● Width: 7mm ● Material: Chrome Steel ● Shield: Metal Shields (ZZ) or Rubber Seals (RS) ● Load Rating: Dynamic: 3.6 kN, Static: 1.55 kN ● Max Speed: 34,000 RPM (for ZZ), lower for RS due to sealing friction ● Usage: Suitable for high-speed, low-noise applications; commonly used in linear motion systems.
20	Threaded Rod	● **Model:** M8 Threaded Rod ● Diameter: 8mm ● Length: 500mm ● Thread pitch 2mm (precise) or 8mm (speed) ● Material: Stainless Steel ● Thread Pitch: 1.25mm (standard for

		M8) ● Tensile Strength: 70,000 to 100,000 psi (depending on material grade) ● Finish: Plain or Zinc-plated for additional corrosion resistance ● Usage: Used in linear motion systems, CNC machines, and load-bearing structures.
21	Flexible Shaft Coupling	● **Model :** Aluminum Flexible Shaft Coupling ● Outer Diameter (OD): 19mm ● Length (L): 25mm ● Material: Aluminium Alloy ● Flexibility: Can accommodate slight angular, parallel, and axial misalignments ● Torque Rating: Up to 2 Nm ● Usage: Connects motor shafts to lead screws or other shafts, reduces misalignment stress.
22	Round Wooden Stick	● **Material :** Hardwood ● Diameter: 8mm (matching the coupling and bearing specifications) ● Length: 500mm ● Finish: Sanded smooth, optionally treated with a clear finish or varnish for durability ● Usage: Can be used as a non-metallic, lightweight alternative for structural or mechanical components.

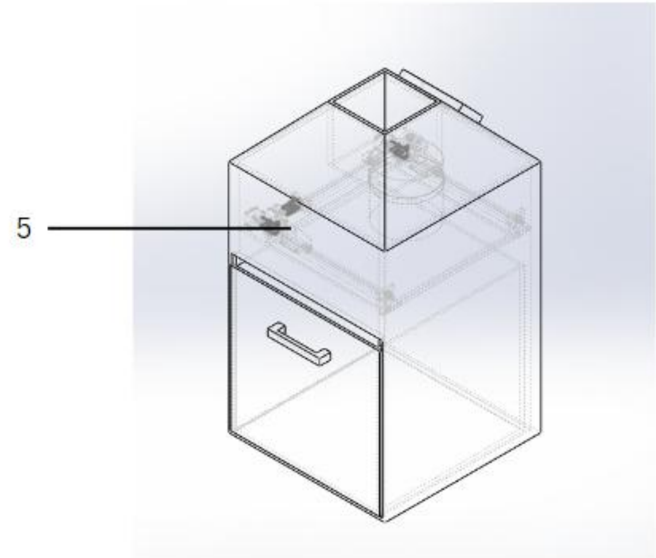
5.2 Embodiment Design

The Automatic Waste Segregation



1	Storage Box
2	Input Hopper
3	User interface section
4	Frame

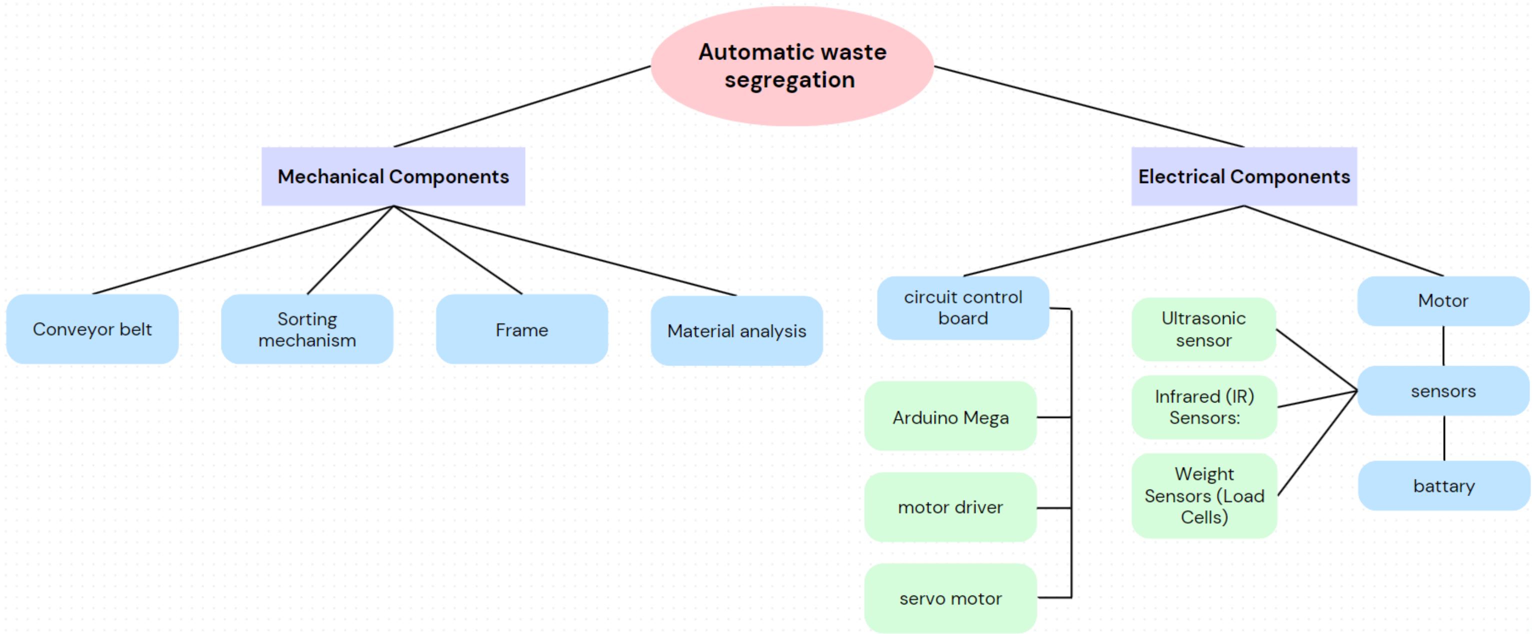
Embodiment Image 1.0



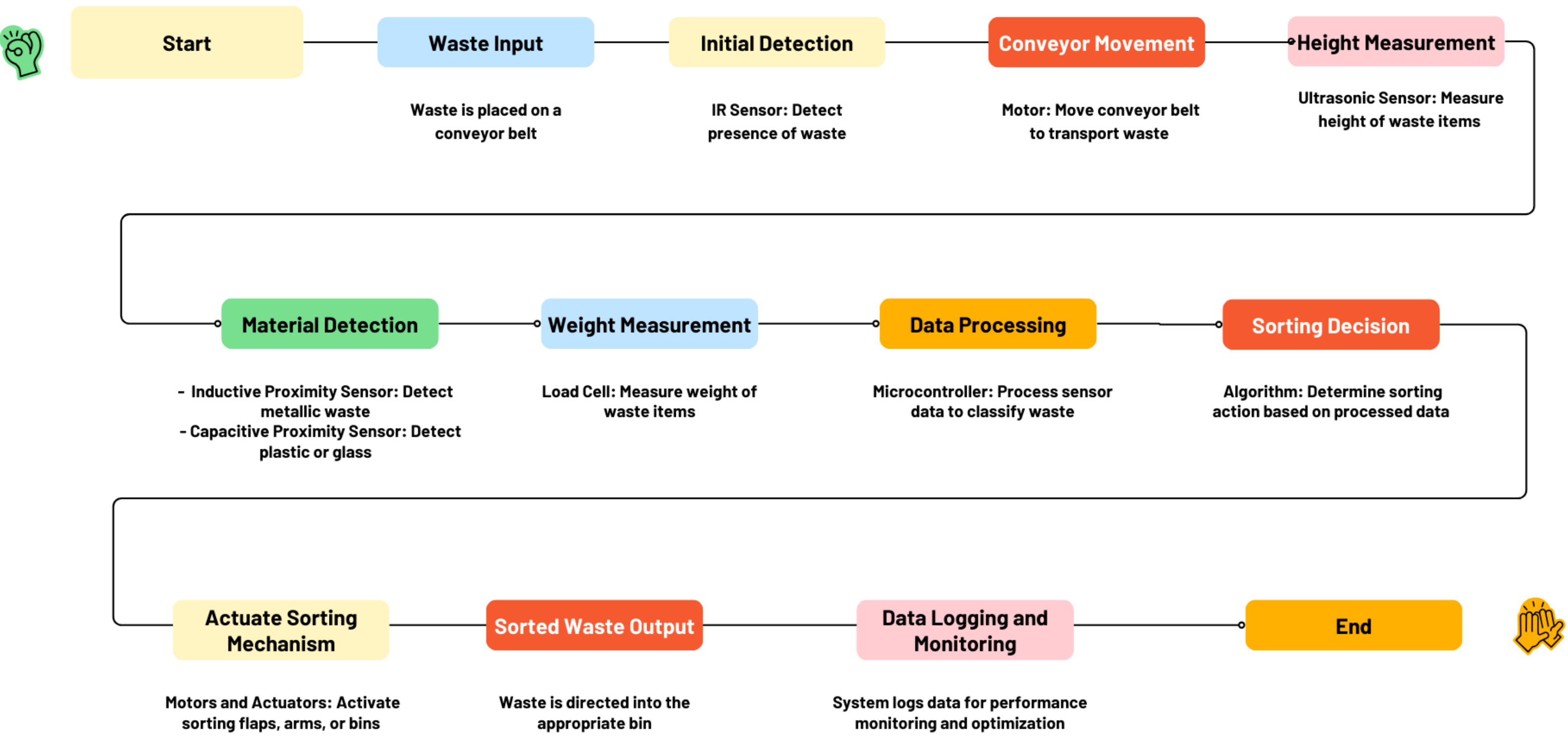
5	Segregate Mechanism

Embodiment Image 1.1

5.2.1 Product dissection



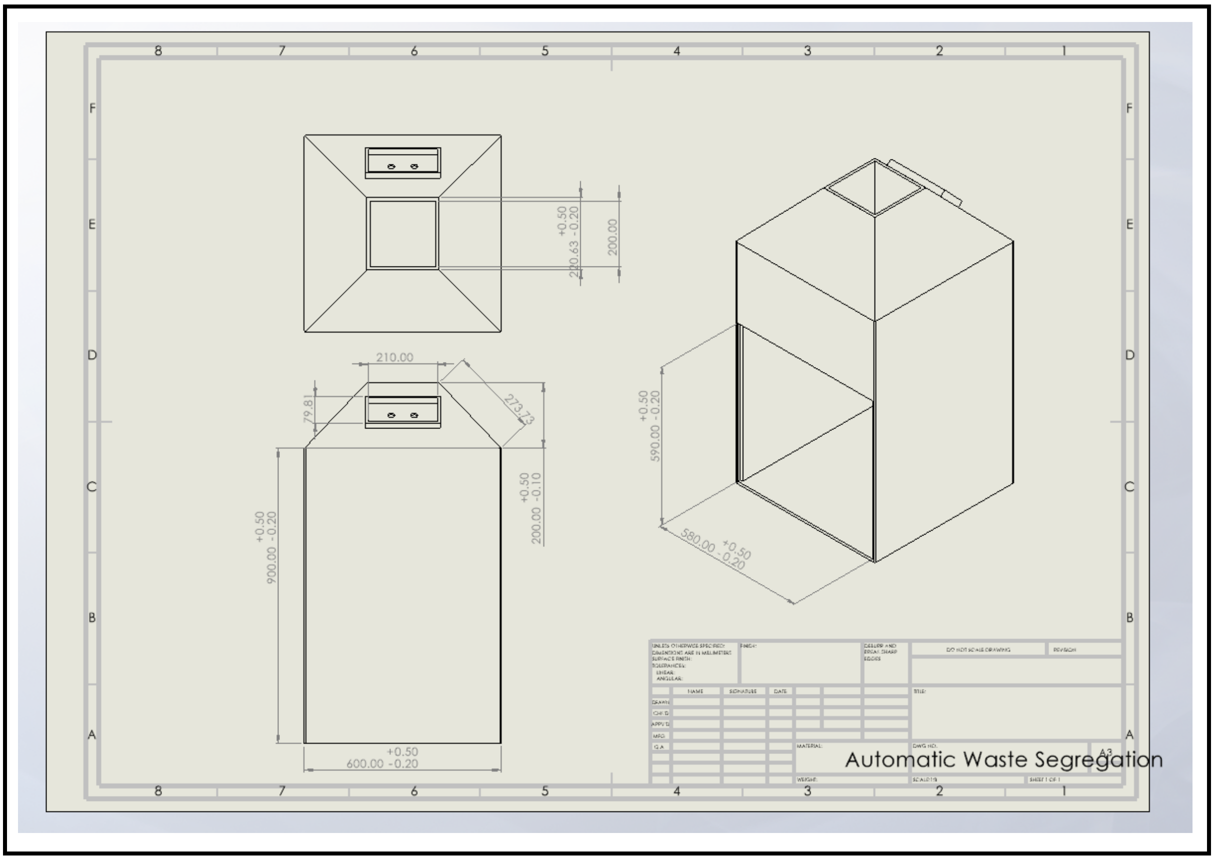
5.2.2 Function structure



5.3 Detail Design

The drawing for Automated Waste Segregation :

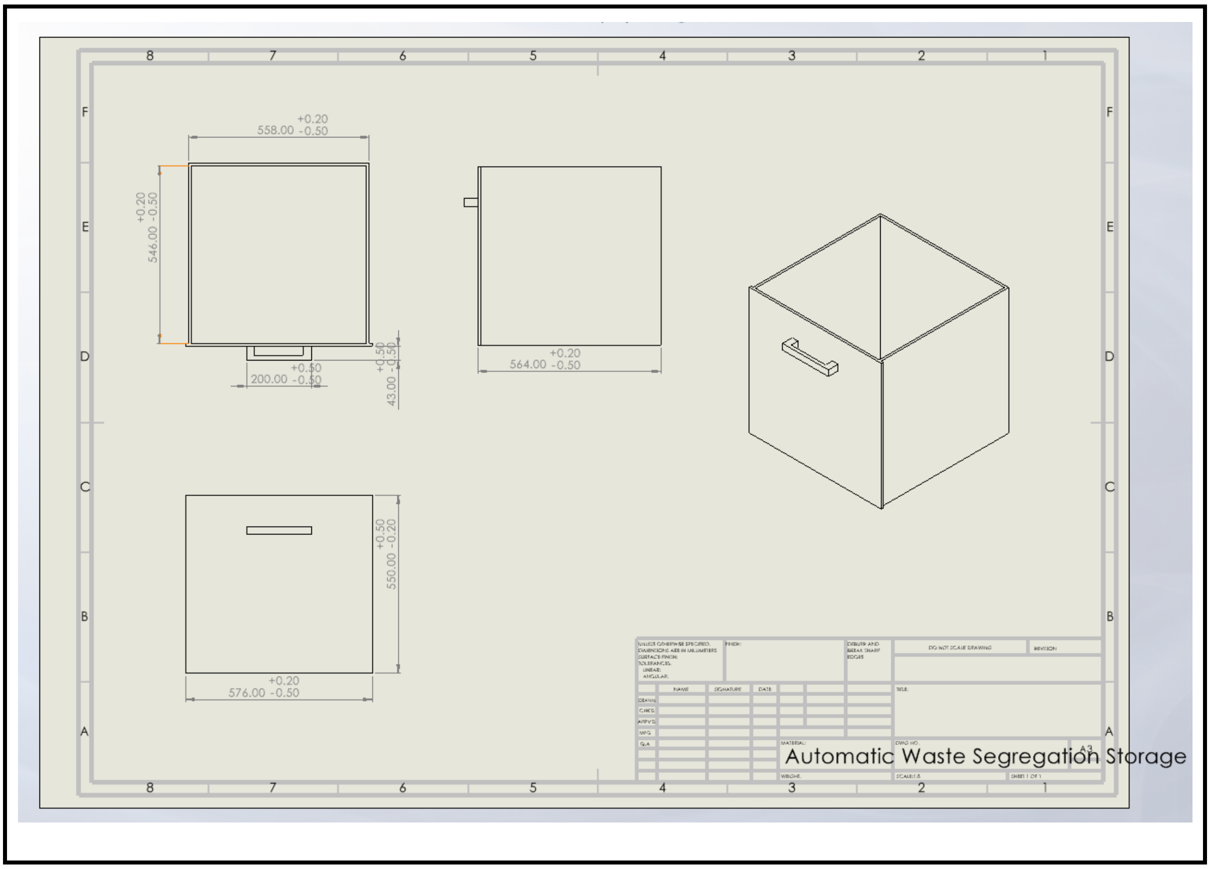
1. Main Frame



Detail Image 1.0

Basically, this is the main frame. The outer body for the automatic waste segregation consists of a storage box, input hopper and user interface section.

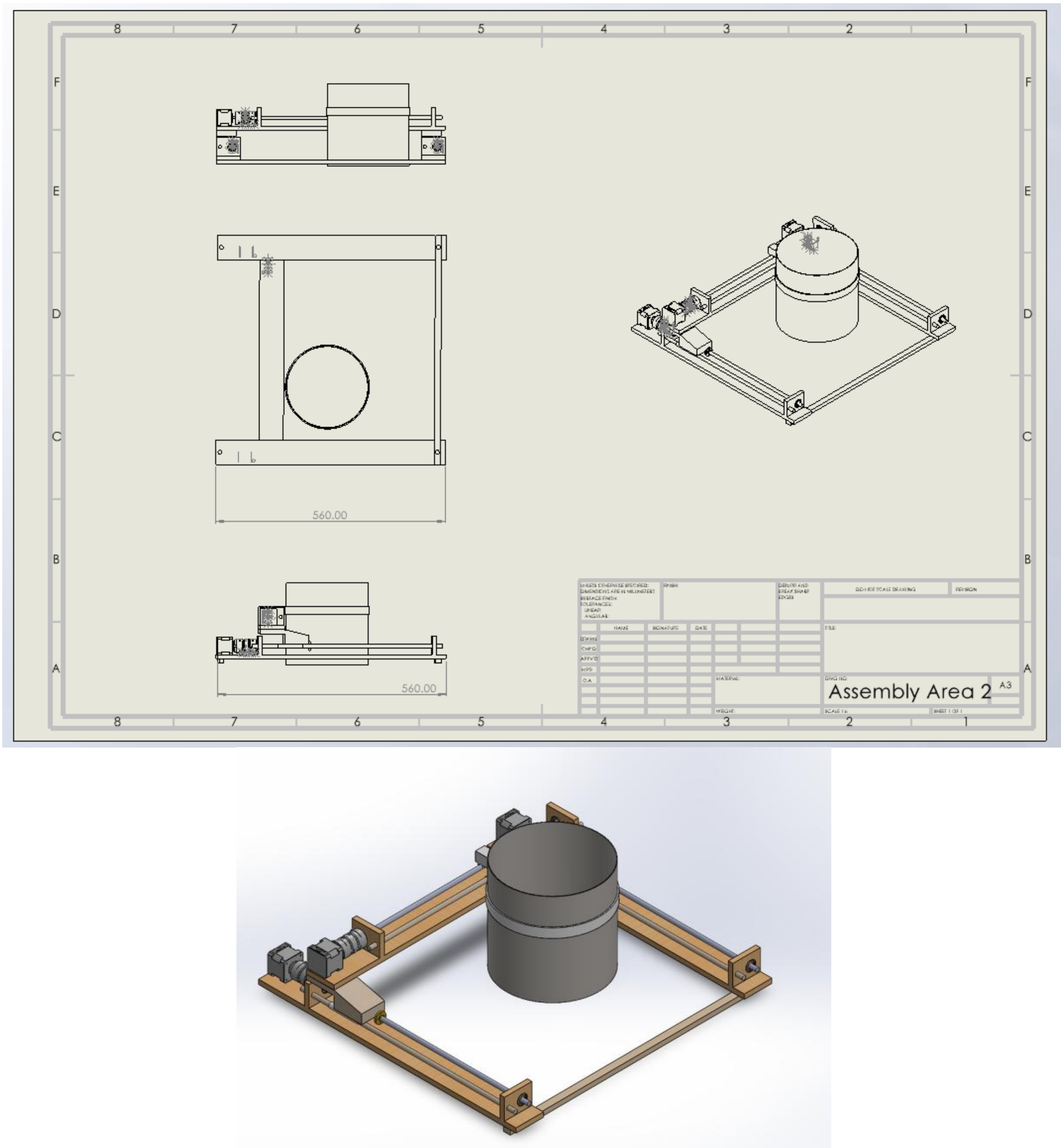
2. Storage Box



Detail Image 1.1

In this part, the storage box will store the recycled waste, in this part,there will be 4 compartments which are compartments for metal, plastic, glass and paper. After the waste identified by the camera, the waste will drop in its compartment.

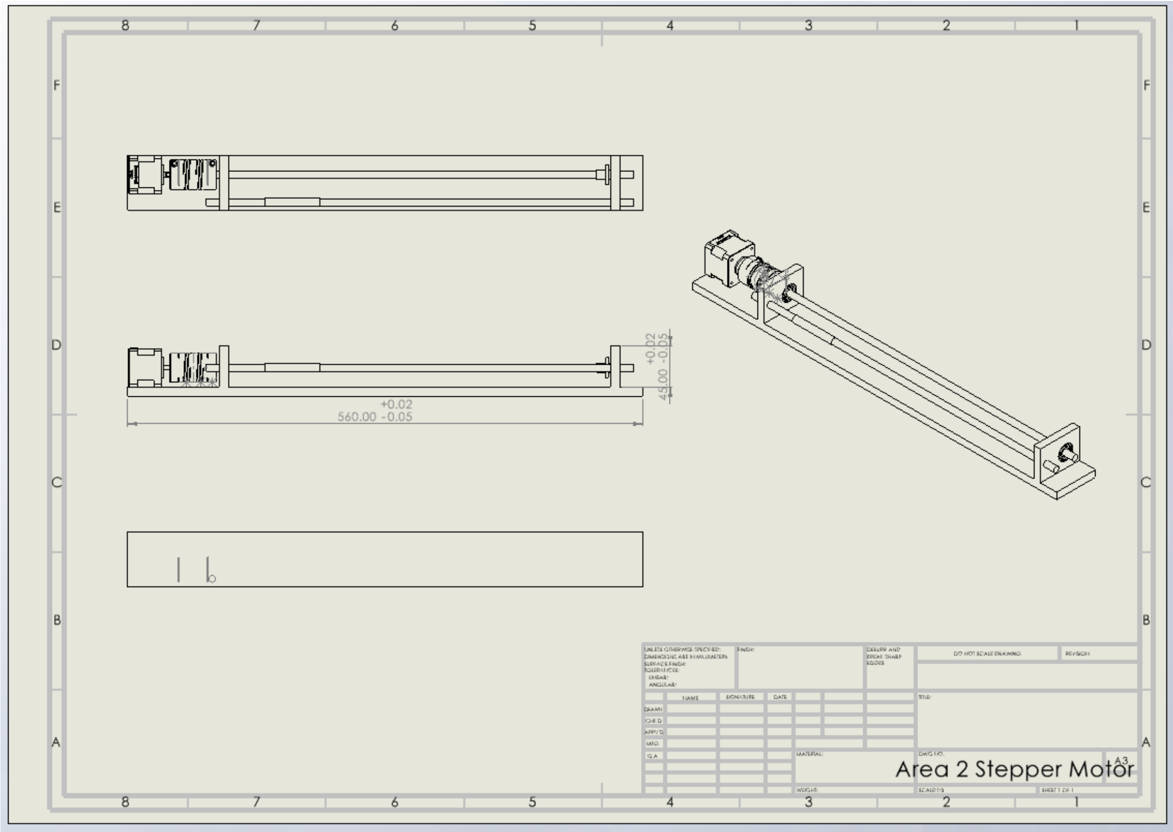
3. Segregate mechanism

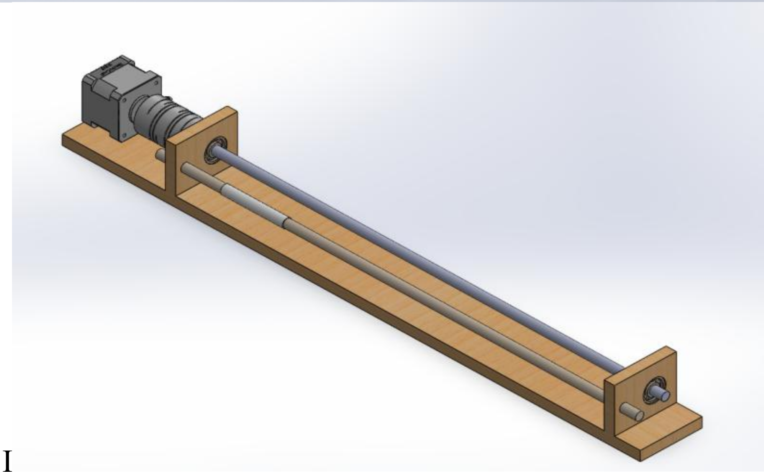


Detail Image 1.3

This is the segregate mechanism part. This part works like a 3D printer, can move x and y axis, after the camera identifies the waste, this mechanism will move to align with its compartment on the storage box.

4. Stepper Motor Platform





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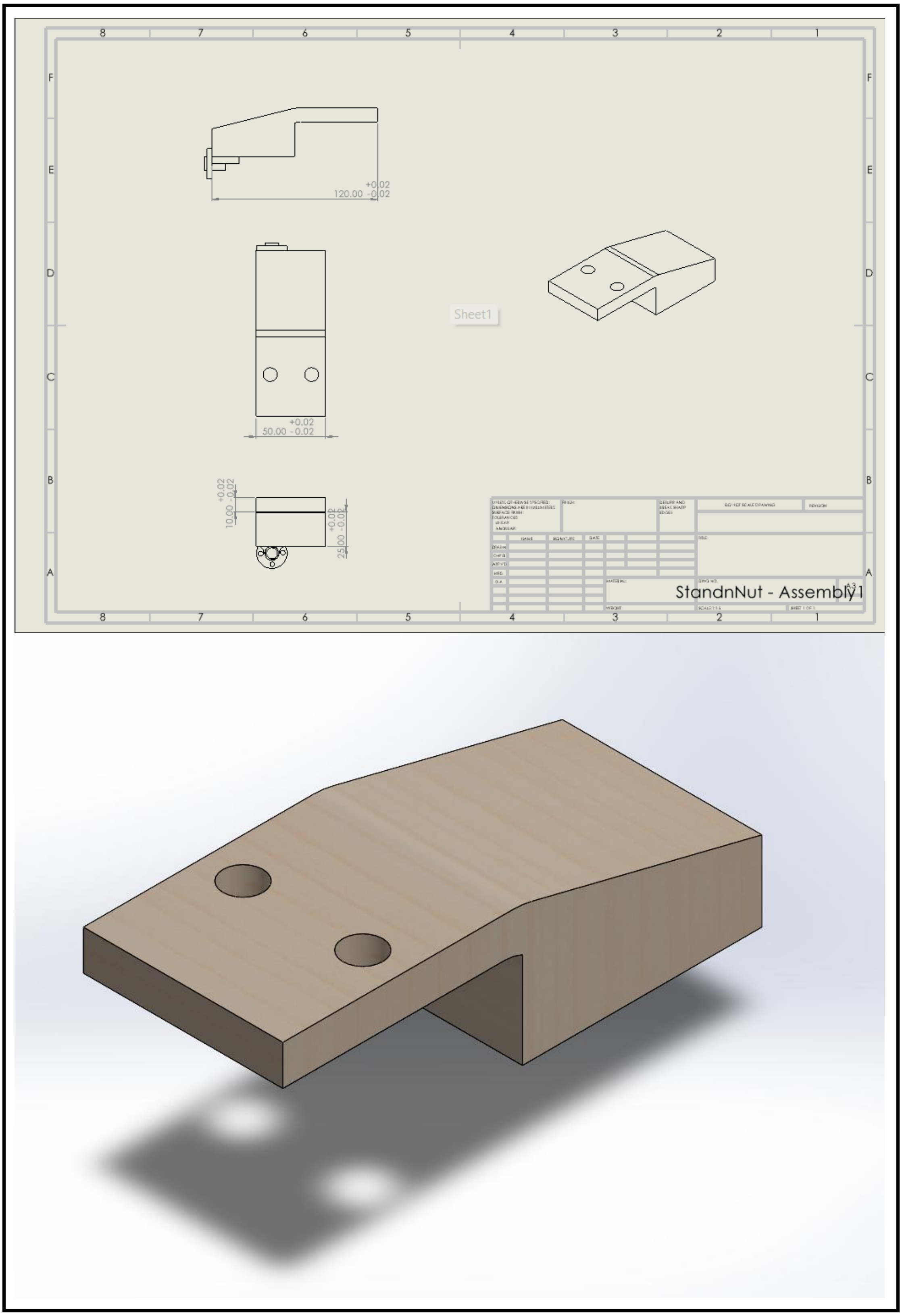
Detail Image 1.4

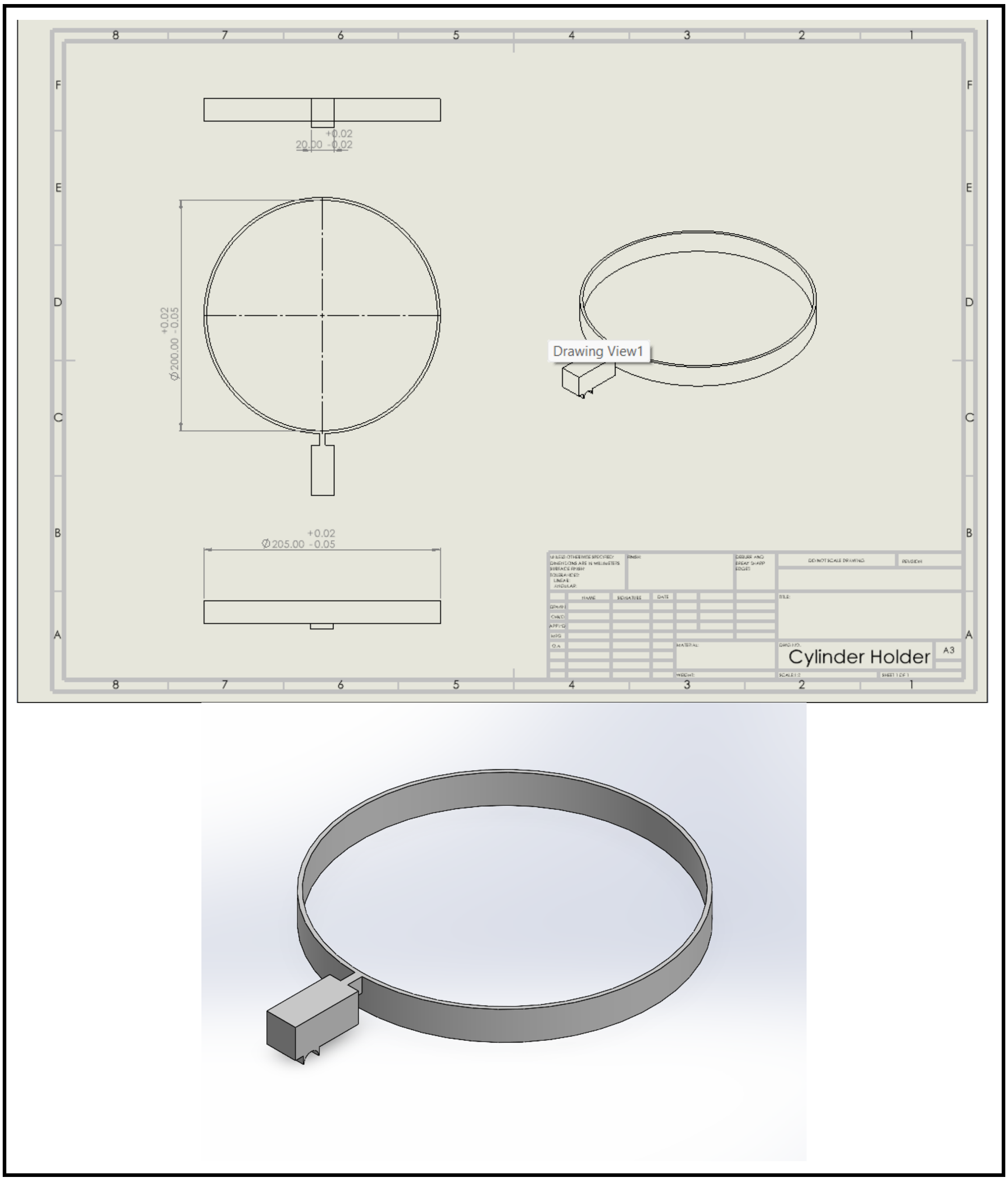
This is the stepper motor platform, Combination of 3 from this part becoming a segregate mechanism.

5. Stepper Motor Platform Holder



Detail Image 1.5

6. Cylinder Holder (Segregate Mechanism)



Detail Image 1.6

For this part, it will hold the cylinder which when the machine receives the waste, the waste will flow from the input hopper to this cylinder. At this cylinder, the identify process will start.

5.4 Design of Manufacture (DFM)

Design for Manufacture (DFM) is an engineering method dedicated to crafting products that enhance the manufacturing process. The aim of DFM is to boost efficiency, reduce production costs, and elevate the quality of the final product.

In our automatic waste segregation project, it has been mandatory to follow the Design for Manufacture (DFM) approach to achieve maximum efficiency for the design and cost implementation of the system. It was possible to move from the use of cumbersome and expensive materials to more economical types without having to lose the effectiveness of the products. For example Storage Box is made of the high plastic material, which was chosen for the fact that it is rather durable and has a good form for molding, which will reduce the cost of manufacturing and production. Thus, the Segregate Mechanism uses standardized constituents to simplify the assembly process and reduce the amount of unique constituents needed. This product is a Stepper Motor Holder made of wooden material; the choice is mainly due to the affordable price and ease of processing, which helps to keep the price low and environmentally friendly. Last but not the least, the Cylinder Holder made out of sturdy steel aims at strength and precision manufacture hence assuming the responsibility of holding the mechanism firmly in operation. These choices and the modifications made to the design guarantee that each component is not only simple and cost-effective to manufacture but also meets our mission to design a dependable waste sorting system. Also, tight tolerances are required when it comes to Segregate Mechanism and Cylinder Holder to guarantee flawless interlocking and eliminate possible misalignments which could negatively impact operations or wear down components. Maintaining both parsimony and efficiency, our DFM approach allows us to propose a solution of the maximal quality and adaptability to various segregation requirements while not going over budget.

5.5 Design of Assembly (DFA)

Design for Assembly (DFA) is an engineering strategy aimed at simplifying the assembly process of products. The core objective of DFA is to streamline the assembly by reducing the number of parts involved, making the components easier to put together, and enhancing overall manufacturing efficiency.

In our project focused on automatic waste segregation, we applied Design for Assembly (DFA) principles to enhance the efficiency and user-friendliness of the system. We designed each component—the Storage Box, Segregation Mechanism, Stepper Motor, Stepper Motor Holder, and Cylinder Holder (Segregation Mechanism)—with simplicity and ease of assembly in mind. The Storage Box is crafted for straightforward integration with the other components, ensuring quick and seamless assembly. The Segregation Mechanism is modular, allowing for independent assembly and easy maintenance or replacement. To further streamline the process, the Stepper Motor Platform and its Holder are designed to fit together effortlessly, minimizing the need for complex adjustments or specialized tools. The Cylinder Holder, essential for the

Segregation Mechanism, is precisely engineered to ensure a snug fit, reducing potential alignment issues and ensuring reliable operation. By carefully selecting materials and designing tolerances for critical areas, we ensured that the assembly process is smooth and efficient. This approach not only simplifies the build and setup of the system but also enhances the overall reliability and performance of our automatic waste segregation solution.

5.6 Costing

No	Item	Shop	Quantity	Cost (RM)
1	Arduino Mega 2560	Shoppee	1	RM 55.90
2	DRV8825 DC CNC Stepper Motor Driver	Shoppee	3	RM 3.30
3	Bipolar Stepper Motor NEMA 17	Shoppee	2	RM 32.00
4	Servo Motor MG996R	Shoppee	1	RM 11.92
5	Ultrasonic sensor HC-SR04	Shoppee	4	RM 2.85
6	Limit switch	Shoppee	1	RM 1.20
7	Wifi module (ESP8266)	Shoppee	1	RM 16.90
8	5mm LED	Shoppee	2	RM 0.79
9	LCD 16x2 (HD44780)	Shoppee	1	RM 11.88
10	Wire	Shoppee	1	RM 9.80
11	Square Hollow Stainless Steel	Shoppee	4	RM 13.50
12	Breadboard	Shoppee	1	RM 5.67
13	Raspberry Pi Camera Module V2	Cytron	1	RM 82.50
14	Power Adapter (barrel jack)	Shoppee	1	RM 13.80
15	TTP223 Capacitive Touch Sensor Module	Shoppee	1	RM 0.77
16	Li-Ion Rechargeable Battery Pack	Shoppee	1	RM 15.20
17	Arduino mega Shield - CNC Shield V3	Shoppee	1	RM 20.90
19	Bearing	Shoppee	6	RM 6.80

20	Threaded Rod	Shoppee	3	RM 25.42
21	Flexible Shaft Coupling	Shoppee	3	RM 2.80
22	Round Wooden Stick	Shoppee	3	RM 10
Total costs				RM 541.99

5.7 Discussion on Safety, Environment, Social Impact and Sustainability Product

When proposing an Automatic Waste Segregation System (AWSS), it is crucial to think about Safety, Environment and Societal Implications since the three elements are among the most important criteria governing the efficiency, sustainability and beneficial effects of the proposed system in society.

5.7.1 Safety

Security is very important and has been incorporated into the AWSS design to enable it to operate efficiently while protecting its users. Safety remains one of the most important aspects in designing and therefore we have several aspects that are put in place in this aspect. First, an emergency shut off mechanism is to be incorporated where it can be accessed through the physical main switch on the machines. This emergency switch is a decided advantage as it immediately shuts down the entire power governing the system in case of any incident. Moreover, we provide the system alarm in the form of light (LED) alerts and sound (buzzer) to compel users to stop using the equipment to prevent possible dangers to the user and the machine. These safety measures further correlate with our AMBITION to use in a safe AWSS that provides safety to its users whilst upholding functionality always.

5.7.2 Environment

Environmental issues therefore come into play as AWSS plans on minimizing landfill waste in addition to advocating for recycling. This means that through grouping recyclable materials well to allow easy practice of recycling, the system assists in sustaining resources and cutting down impacts of waste disposal on the environment. Some of the developments including energy efficiency and new energy sources reduce the emission of carbon and help upgrade the waste disposal practices. This can be a specific air or water quality that measures indicate that waste segregation activities are not affecting the environment adversely.

5.7.3 Social Impact

Community Health, Resources, and social issues are the last aspects that are affected by AWSS in the lives of people across the societies. In a way that boosts the quality environment and fights pollution, AWSS reduces various health hazards and effects associated with improper disposal of wastes in the communities. Optimization in waste processing processes contributes to the enhancement of resource recycling practices bringing minimal impacts on the extraction of raw materials. In the same way, that will generate employment chances where some of them are in waste management and recycling industries where it will help the economy and also assist in achieving equal chances of accessing the services by all the people regardless of their status.

5.7.4 Sustainability

In this study, the sustainability of AWSS is measured by the ability of the system to optimally use resources within its production process, and the least environmental degradation within its lifespan with reference to sustainability. This entails choosing long-lasting, sustainable and non-hazardous materials into the to be developed facility, incorporating green energy systems like photovoltaic system and regular practice of source reduction on wastes through segregation and recycling. Doing Life Cycle Assessments (LCAs) before an item is manufactured and produced, reduces its effect on the environment when it is being manufactured, used, and even disposed of. Thus, it is important for AWSS to follow environmental measures and utilise resources in such a way as to cause less or no harm to the environment, due care and maintenance for the equipment and following sustainable waste disposal techniques.

Consequently, factors such as safety, sustainability, environment and social effect should be included into the AWSS in order to safeguard operational safety, environmental friendliness as well as inculcate the positive impacts of safety on the society such as improved quality of life due to less incidences of injuries, preservation of natural resources as well as provision of jobs. These aspects in unison strengthen the system as well as its impact on the overall benefits to local communities, and larger environmental concerns.

5.8 Gantt Chart 2